

Statistics

Probability distribution

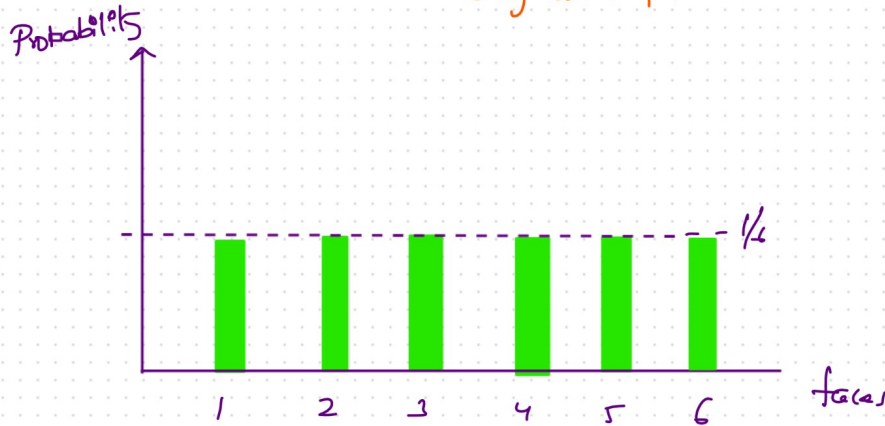
PMF (probability mass function)

→ PMF is a function

To answer: "what is the chance of X happening exactly?"

$$\sum P(n) = 1 \quad (\text{sum of all prob} = 1)$$

only works for discrete outcomes



$$S = \{1, 2, 3, 4, 5, 6\}$$

$$1 : 1/6$$

$$2 : 1/6$$

$$3 : 1/6$$

$$4 : 1/6$$

$$5 : 1/6$$

$$6 : 1/6$$

CPF (Cumulative Distribution function)

→ CPF works for both discrete & continuous data

→ It gives the probability that X is $\leq$ a value

why?

To answer: what's the chance of X being less than or equal to 5?

ex: what is the prob of rolling ≤ 5 on a dice?

$$P(1) = 1/6$$

$$P(2) = 1/6$$

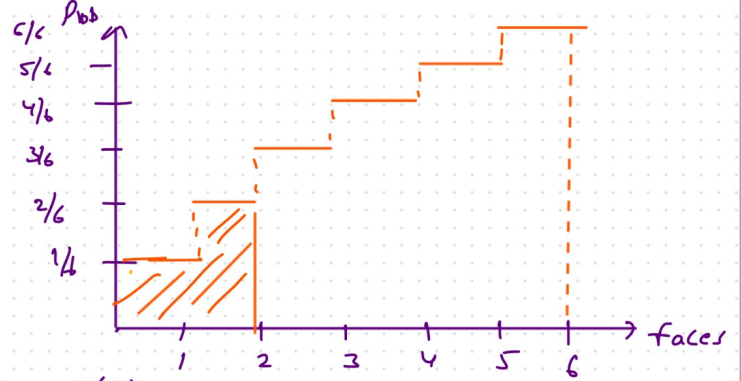
⋮

$$P(6) = 1/6$$

$$P(\leq 2) = P(1) + P(2)$$

$$P(\leq 5) = P(1) + P(2) + P(3) + P(4) + P(5)$$

Cdf (≤ 2)



$$P(0) = 0$$

$$P(\leq 1) = \text{pmf}(0) + \text{pmf}(1) \\ = 0 + 1/6 = 1/6$$

$$P(\leq 2) = \text{pmf}(0) + \text{pmf}(1) + \text{pmf}(2) \\ = \frac{2}{6}$$

$$P(\leq 6) = \text{pmf}(0) + \dots + \text{pmf}(6) \\ = \frac{6}{6} = 1$$

→ cdf always starts at 0 & ends at 1

→ discrete data cdf is a step function

→ for continuous data, cdf is smooth curve

discrete → pmf
→ cdf

continuous → ~~pdf~~ pmf?
→ cdf

↓
 $P(x)$

Q) why not pmf for continuous data?

→ what is the probability of a student height is 160 cm

[150 cm, 170 cm]
?

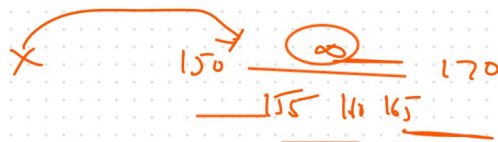
150.1 150.4 150.5
160

$$P(160) = \frac{1}{\infty} \\ = \underline{0}$$

$$P(160) \rightarrow 0 \rightarrow 3\% \\ 40\%$$

0.40%
0.35%
0.75%

→ what's the probability a student height is between 166 cm - 170 cm ?
range

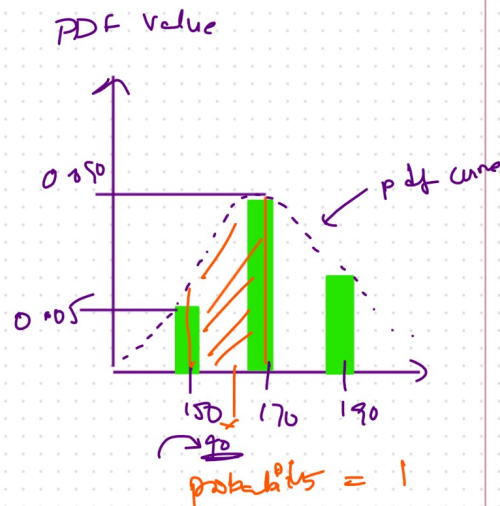


PDF:

- PDF is for continuous data like (height, weight, temp)
- It shows the relative likelihood of a range of values (not exact point)
- Continuous probability are meaningful over intervals

PDF → as density (Not probability)

→ The height of the pdf curve at a point tells you how concentrated probabilities are around that curve



Higher pdf value = more likely to find values nearby

Step 1 Collect the Data

Height (cm) of 20 people

$n = 20$

min - 154

max - 179

{ 162, 168, 171, 154, 175, 178, 165, 169, 172, 160, 167, 173, 170,

176, 164, 179, 166, 170, 174, 168 }

step 2 : Create a histogram

Bin's → frequency

height (cm)	frequency
150 - 155	1
155 - 160	0
160 - 165	3
165 - 170	6
170 - 175	6
175 - 180	4

$n\text{-bin} = 6$
width = 5

Step 3: Normalize Histogram to get PDF

$$PDF = \frac{\text{frequency of bin}}{\text{total observation} \times \text{bin width}}$$

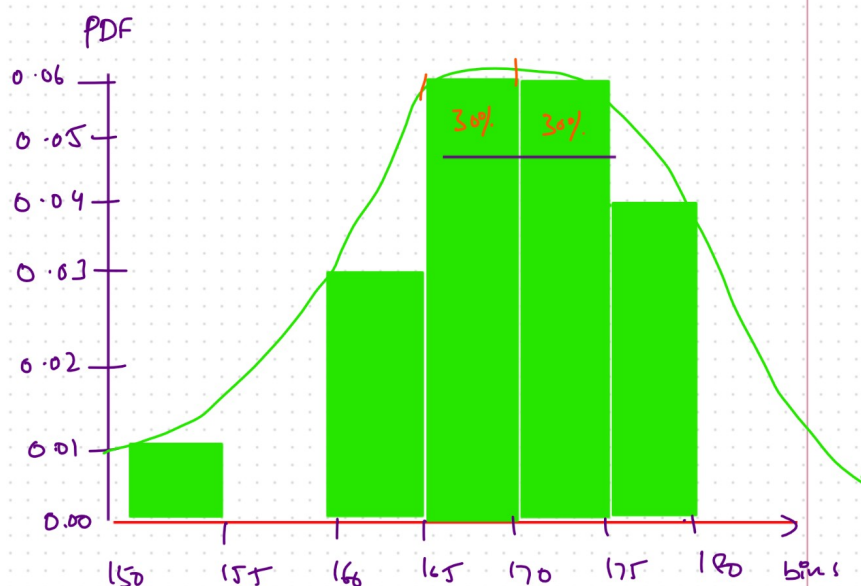
$$= \frac{\text{frequency of bin}}{\text{total observation} \times \text{bin width}}$$

$$\frac{1}{20 \times 5} = \frac{0.05}{5} = 0.01$$

$$\frac{1}{20 \times 5} = \frac{1}{100} = 0.01$$

$$\frac{9}{20 \times 5} = \frac{9 \times m}{b \times n}$$

Height (cm)	frequency	PDF
150 - 155	1	$\frac{1}{20 \times 5} = 0.01$
155 - 160	0	0
160 - 165	3	0.03
165 - 170	6	0.06
170 - 175	6	0.06
175 - 180	4	0.04



(Q) $P(165 \leq \text{height} \leq 170)$

step 4 : Calculate area under PDF curve

→ Identify bins involved : 165 - 170

$$\text{Area} = \text{PDF value} \times \text{Bin width}$$

$$= 0.06 \times 5 = 0.30 \text{ (30\% chance)}$$

Verification

Actual data point in 165 - 170 cm : 6 people

$$\text{probability} = \frac{6}{20} = 0.30\%$$

example

$$P(\text{height} \leq 170)$$

1) Add areas upto 170 cm:

$$150-155 = 0.01 \times 5 = 0.05$$

$$155-160 = 0 \times 5 = 0$$

$$160-165 = 0.03 \times 5 = 0.15$$

$$165-170 = 0.06 \times 5 = 0.30$$

~~170-175~~

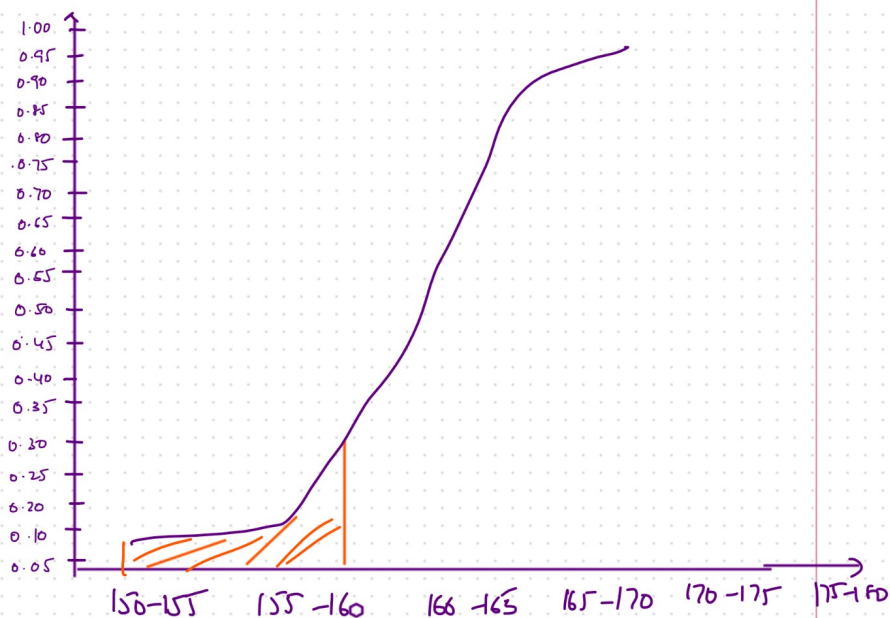
$$\text{total} = 0.05 + 0 + 0.15 + 0.30$$

$$= 0.50 \rightarrow 50\%$$

$$\text{verification} = \frac{10}{20} = 0.5 = 50\%$$

CDF

	Height (cm)	frequency	pdf	Probability
A	150-155	1	0.01	0.05
B	155-160	0	0	0
C	160-165	3	0.03	0.15
D	165-170	6	0.06	0.30
E	170-175	6	0.06	0.30
F	175-180	4	0.04	0.20



CDF

$$A \rightarrow 0.05$$

$$B \rightarrow 0 + 0.05 \rightarrow 0.05$$

$$C \rightarrow 0.05 + 0.15 \rightarrow 0.20$$

$$D \rightarrow 0.20 + 0.30 \rightarrow 0.5$$

$$E \rightarrow 0.50 + 0.30 \rightarrow 0.8$$

$$F \rightarrow 0.8 + 0.20 \rightarrow 1$$

$$P(a \leq x \leq b) = CDF(b) - CDF(a)$$

$$\begin{aligned}
 P(165 \leq x \leq 170) &= CDF(170) - CDF(165) \\
 &= P(x \leq 170) - P(x \leq 165)
 \end{aligned}$$

Histogram

100 110 120 130 140 ← flattened / uniformly

distributed

100 - 110 - 1

110 - 120 - 1

- 1

- 1

